FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 25

By - Dr. Puneet Sharma Sir



Recap *of previous lecture*

SET THEORY



Topics to be covered

" RECURRENCE RELATION "

(RECURSIVE RELATION)



Eg Let us Consider a Sequence;

1, 3, 5, 7, 9, 11 - - - - - - -

then we can represent it as

(M-I) By G-formula; $\boxed{a_n = (2n-1)}$, $n \in \mathbb{N}$

(M-II) By R-Relation; $\boxed{a_n = a_{n-1} + 2}$, $a_0 = 1$

$\Rightarrow$ 1, 3, 5, 7, 9, - - - -

Note: $a_0, a_1, a_2, a_3, a_4 \dots \to a_{n-1}, a_n$

To find G-formula for a_n with the help of RR:

$\boxed{a_n - a_{n-1} = 2}$, $a_0 = 1$

$a_1 - a_0 = 2$

$a_2 - a_1 = 2$

$a_3 - a_2 = 2$

$a_4 - a_3 = 2$

$\vdots$

$a_{n-1} - a_{n-2} = 2$

$a_n - a_{n-1} = 2$

n Pairs

Adding: $a_n - a_0 = 2n$

$a_n = 2n + 1$, $n \in W$

$\Downarrow$

$\boxed{a_n = 2n - 1}$, $n \in \mathbb{N}$

eg Let us Consider another ex

4, 5, 7, 10, 14 - - - - - -

then it may be represented as

(M-I) $0+4,\ 1+4,\ 3+4,\ 6+4,\ 10+4, \ldots\ldots$

w.k.that $\Sigma(n) = \frac{n(n+1)}{2}$ $\Sigma(1) = 1$, $\Sigma(2) = 3$, $\Sigma(3) = 6$

$\Rightarrow 0+4,\ \Sigma(1)+4,\ \Sigma(2)+4,\ \Sigma(3)+4,\ \Sigma(4)+4, \ldots, \Sigma(n)+4$

ie $\boxed{a_n = \frac{n(n+1)}{2} + 4}$

(M-II) $\boxed{a_n = a_{n-1} + n}$; $a_0 = 4$

$\Rightarrow a_0, a_1, a_2, a_3, a_4 - - - -$

or $4,\ 4+1,\ 5+2,\ 7+3,\ 10+4 - - - -$

ie $4, 5, 7, 10, 14 - - - - -$

↑ this is called R.R of given sequence.

General formula with the help of RR →

$a_n - a_{n-1} = n$

$a_1 - a_0 = 1$

$a_2 - a_1 = 2$

$a_3 - a_2 = 3$

$\vdots$

$a_n - a_{n-1} = n$

Adding all these we get,

$a_n - a_0 = 1+2+3+ \cdots + n$

$\boxed{a_n = \frac{n(n+1)}{2} + 4}$

Defⁿ:- To write n^{th} term of any sequence with the help of previous terms (with some initial conditions) is called Recurrence Relation.

Solⁿ of RR → means to find the value of a_n in terms of G. formula

There are 3 methods to sol R.R ;

(M-I) Iterative Method (Not in syllabus)

(M-II) Generating function Method. (only for CS/IT)

(M-III) Characteristic Root Method. (≃ Based on CF + PI Concept)

order of RR :- The difference b/n Highest & lowest subscript is called order

Degree of RR :- Highest exponent of a_n occurs in RR is called it's Degree

Linear RR :- if Degree of RR = 1 $\Rightarrow$ it is called Linear RR.

Non Linear RR :- if Degree of RR $\neq$ 1 $\Rightarrow$ $\therefore$ Non linear RR

Homogeneous RR :- If RHS of RR is Zero ie No term of 'n' occurs independently in RR then it is called Homog R.R.

eg (1) $a_{n+3} - 5a_{n+2} + 2a_n + 4a_{n-1} = 0 \Rightarrow$ order = 4, Deg = 1, Linear, Homog

eg (2) $a_n^3 - 2a_{n-1} + 3a_{n-2} = 7^n \Rightarrow$ order = 2, Deg = 3, Non Linear, Non Homog

eg (3) $a_n - a_{n-1} = n$ $\therefore$ order = 1, Deg = 1, Linear, Non Homog.

FIBONACCI SEQUENCE

Consider the seq. 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

it's RR is $a_n = a_{n-1} + a_{n-2}$; $a_0 = 1, a_1 = 1$

& it's G. term is $a_n = \left[\dfrac{\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}}{\sqrt{5}}\right]$; $n \in W$

or $a_n = \dfrac{\left(\frac{1+\sqrt{5}}{2}\right)^{n} - \left(\frac{1-\sqrt{5}}{2}\right)^{n}}{\sqrt{5}}$, $n \in N$

(HW) $a_n - a_{n-1} - a_{n-2} = 0$; $a_0 = 1, a_1 = 1$ — (1)

$\Rightarrow x^2 - x - 1 = 0 \Rightarrow x = \dfrac{1 \pm \sqrt{5}}{2} = \alpha_1, \alpha_2$

$a_n = C_1 \alpha_1^n + C_2 \alpha_2^n$ — (2)

using B.Cond^n, $C_1 = \dfrac{\alpha_1}{\sqrt{5}}$, $C_2 = \dfrac{-\alpha_2}{\sqrt{5}}$

$a_n = \dfrac{\alpha_1^{n+1}}{\sqrt{5}} - \dfrac{\alpha_2^{n+1}}{\sqrt{5}}$, $n \in W$

Linear, Homog RR of nth order with Constant Coeffs →

Eg: Solve; $a_n = 5a_{n-1} - 6a_{n-2}$; $a_0 = 1$, $a_1 = 4$, $n \geq 2$ — (1)

Sol: $a_n - 5a_{n-1} + 6a_{n-2} = 0$

order = 2 ⇒ Linear, Homog, Const Coeffs

$\alpha^2 - 5\alpha^1 + 6\alpha^0 = 0$

$\alpha^2 - 5\alpha + 6 = 0 \Rightarrow \alpha = 2, 3$

Hence Sol of (1) is $a_n = C_1 2^n + C_2 3^n$ — (2)

Using $a_0 = 1 \Rightarrow 1 = C_1 + C_2$ — (3)

Using $a_1 = 4 \Rightarrow 4 = 2C_1 + 3C_2$ — (4)

Solving (3) & (4); $C_1 = -1$, $C_2 = 2$

Hence Sol of RR (1) is $a_n = 2(3^n) - (2^n)$

Cross check: $a_0 = 1$, $a_1 = 4$, $a_2 = 14$, $a_3 = 46$, ----

ie Seq is 1, 4, 14, 46, ------


Thank
YOU